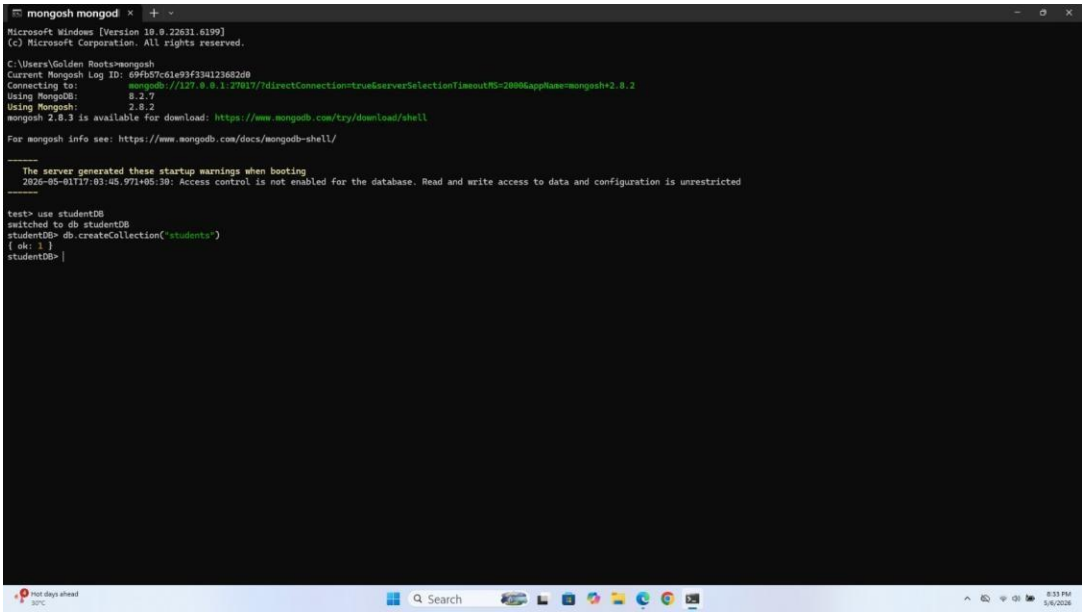
MongoDB CRUD Operations Using Mongosh

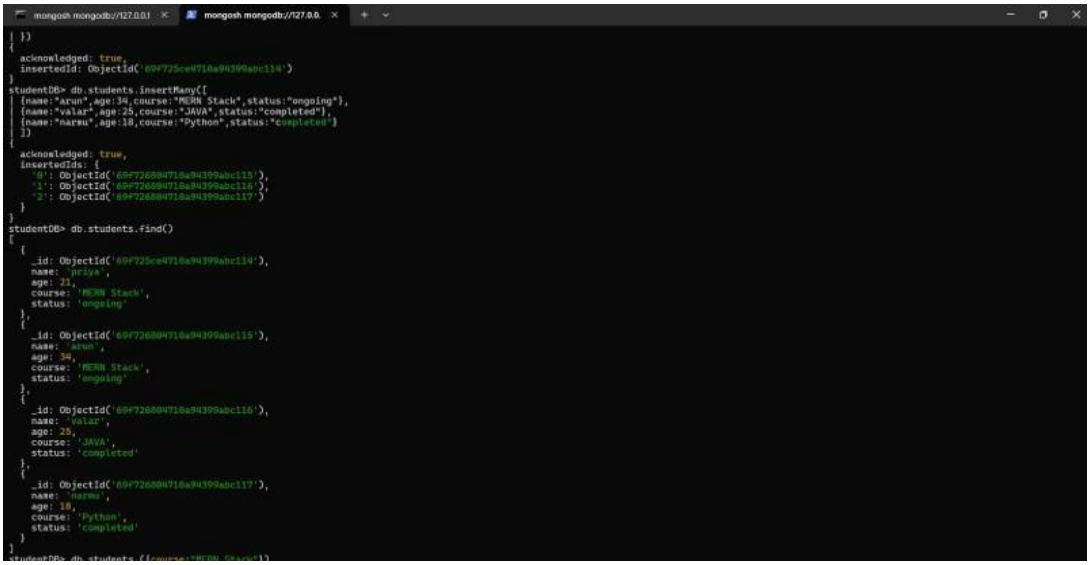
1. Database & Collection Creation

Creation of Database and Collection with Initial Data Insertion



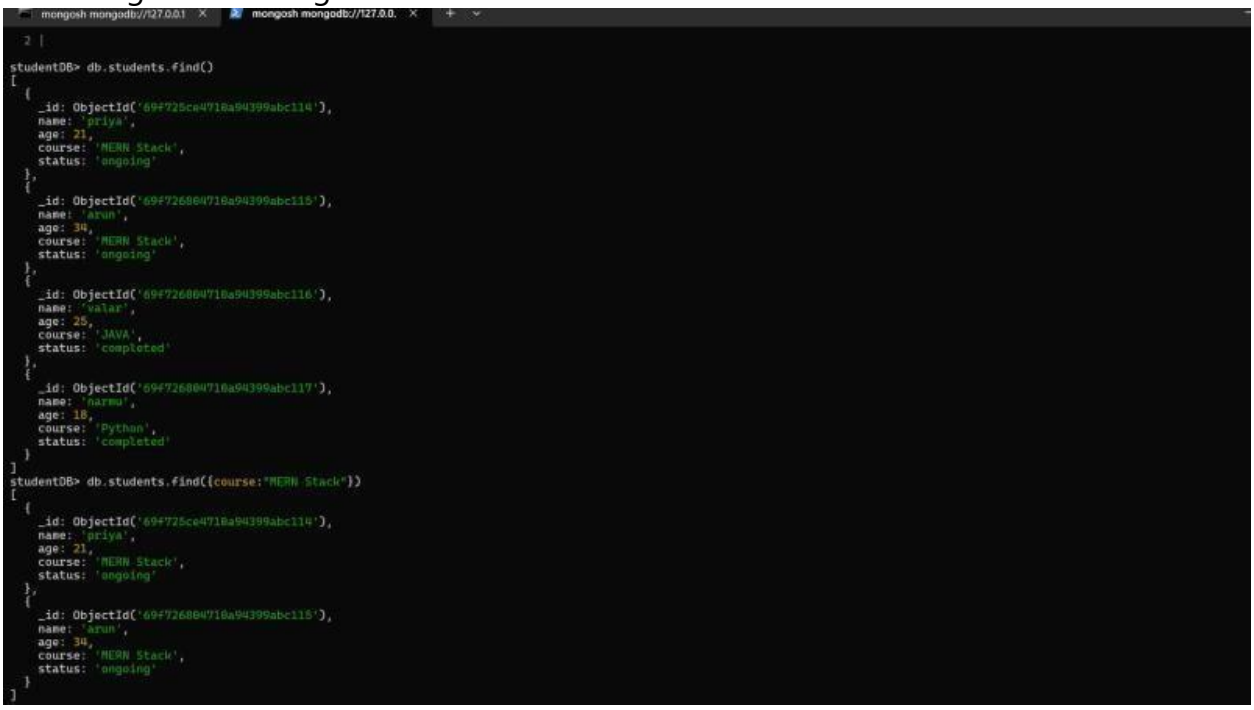
2. Insert Operations

Adding Single and Multiple Documents into the Collection



3. Read Operations

Retrieving and Filtering Data from the Collection



```
mongosh mongodb://127.0.0.1 ×  mongosh mongodb://127.0.0. ×  +

  2 |

studentDB> db.students.find()
[
  {
    _id: ObjectId('69f725ce4718a94399abc114'),
    name: 'priya',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f726804718a94399abc115'),
    name: 'arun',
    age: 34,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f726804718a94399abc116'),
    name: 'valar',
    age: 25,
    course: 'JAVA',
    status: 'completed'
  },
  {
    _id: ObjectId('69f726804718a94399abc117'),
    name: 'narmu',
    age: 18,
    course: 'Python',
    status: 'completed'
  }
]
studentDB> db.students.find({course:"MERN Stack"})
[
  {
    _id: ObjectId('69f725ce4718a94399abc114'),
    name: 'priya',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f726804718a94399abc115'),
    name: 'arun',
    age: 34,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
```

4. Update Operations

Modifying Existing Documents in the Collection

```
studentDB> db.students.updateOne({name:"arun"},
| {$set:{status:"completed"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.updateMany(
|   { course: "MERN Stack" },
|   { $set: { status: "inprogress" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

5. Delete Operations

Removing Documents Based on Conditions

```
studentDB> db.students.deleteOne({ name: "priya" })
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.deleteMany({})
{ acknowledged: true, deletedCount: 3 }
studentDB>  db.students.insertMany([
| {name:"arun",age:34,course:"MERN Stack",status:"ongoing"},
| {name:"valar",age:25,course:"JAVA",status:"completed"},
| {name:"narmu",age:18,course:"Python",status:"completed"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f729234718a94399abc118'),
    '1': ObjectId('69f729234718a94399abc119'),
    '2': ObjectId('69f729234718a94399abc11a')
  }
}
```

6. Query Operators

Applying Conditional Queries for Data Filtering

```
studentDB> db.students.find({ course: { $in: ["Java", "Python"] } })
[
  {
    _id: ObjectId('69f729234710a94399abc11a'),
    name: 'narmu',
    age: 18,
    course: 'Python',
    status: 'completed'
  }
]
studentDB> db.students.find({
|   $and: [
|     { course: "MERN Stack" },
|     { status: "ongoing" }
|   ]
| })
[
  {
    _id: ObjectId('69f729234710a94399abc118'),
    name: 'arun',
    age: 34,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
studentDB> db.students.find({
|   $or: [
|     { course: "Java" },
|     { age: { $lt: 21 } }
|   ]
| })
[
  {
    _id: ObjectId('69f729234710a94399abc11a'),
    name: 'narmu',
    age: 18,
    course: 'Python',
    status: 'completed'
  }
]
```

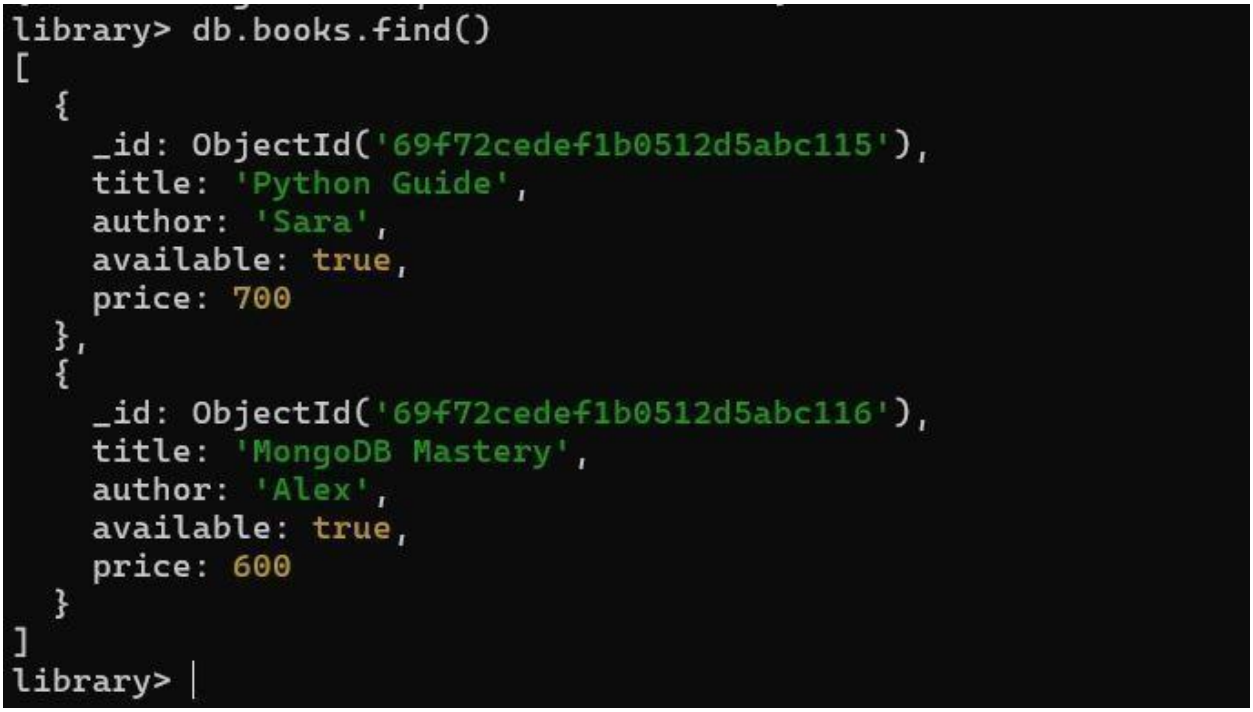
```
studentDB> db.students.find({ age: { $exists: true } })
[
  {
    _id: ObjectId('69f729234710a94399abc118'),
    name: 'arun',
    age: 34,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f729234710a94399abc119'),
    name: 'valar',
    age: 25,
    course: 'JAVA',
    status: 'completed'
  },
  {
    _id: ObjectId('69f729234710a94399abc11a'),
    name: 'narmu',
    age: 18,
    course: 'Python',
    status: 'completed'
  }
]
```

7. Use Case: E-Commerce Real-World Application of MongoDB Operations

Real-World Application of MongoDB Operations

```
test> use library
switched to db library
library> db.createCollection("books")
{ ok: 1 }
library> db.books.insertMany([
| { title: "Java Basics", author: "John", available: true, price: 500 },
| { title: "Python Guide", author: "Sara", available: false, price: 700 },
| { title: "MongoDB Mastery", author: "Alex", available: true, price: 600 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f72cedef1b0512d5abc114'),
    '1': ObjectId('69f72cedef1b0512d5abc115'),
    '2': ObjectId('69f72cedef1b0512d5abc116')
  }
}
library> db.books.find({ available: true })
[
  {
    _id: ObjectId('69f72cedef1b0512d5abc114'),
    title: 'Java Basics',
    author: 'John',
    available: true,
    price: 500
  },
  {
    _id: ObjectId('69f72cedef1b0512d5abc116'),
    title: 'MongoDB Mastery',
    author: 'Alex',
    available: true,
    price: 600
  }
]
```

```
library> db.books.updateOne(
| { title: "Python Guide" },
| { $set: { available: true }
| })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.deleteOne({ title: "Java Basics" })
{ acknowledged: true, deletedCount: 1 }
```



Conclusion:

MongoDB CRUD operations were successfully implemented using mongosh. This exercise helped in understanding how to create, manage, and manipulate data efficiently in a NoSQL database environment.